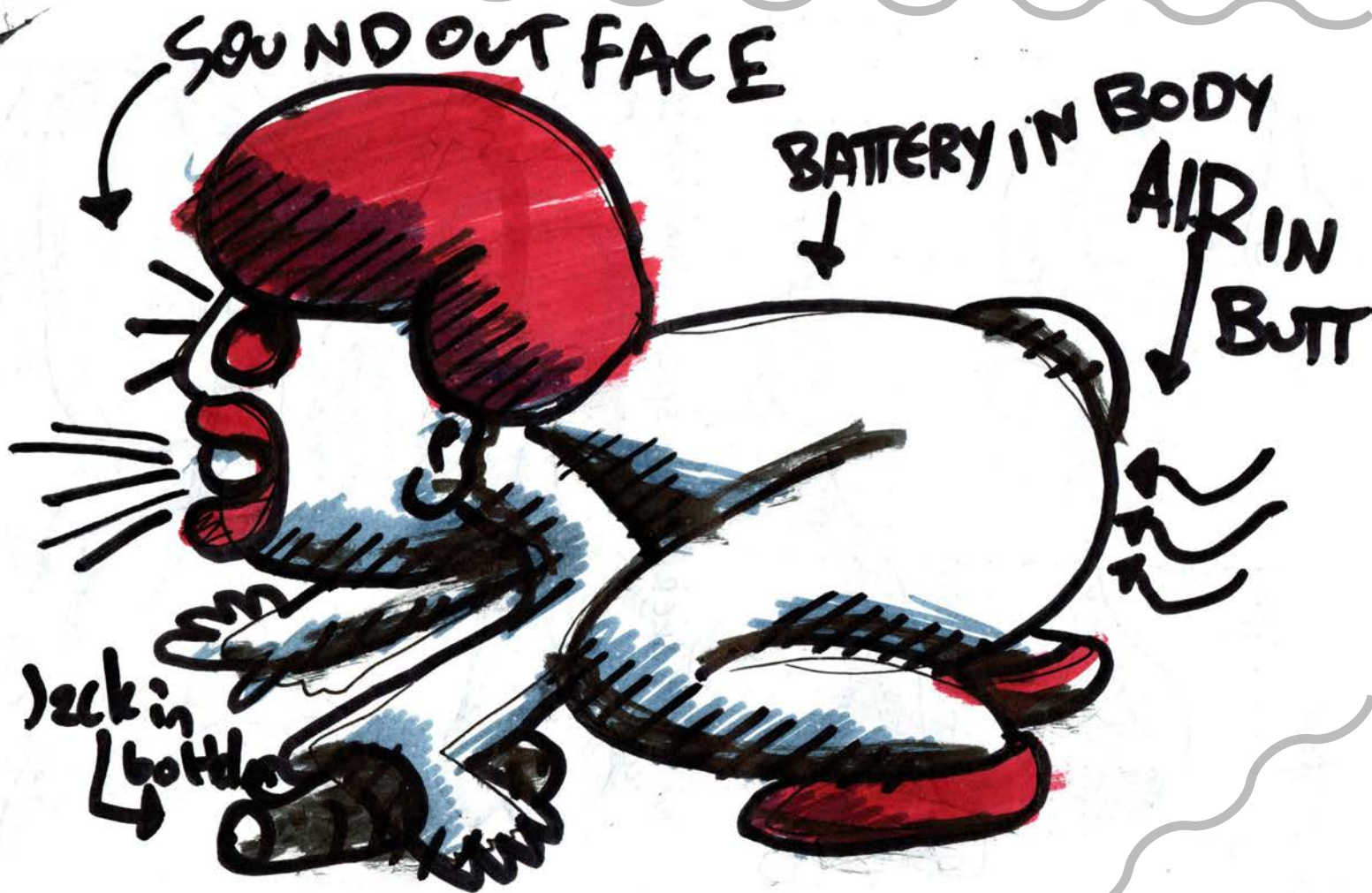
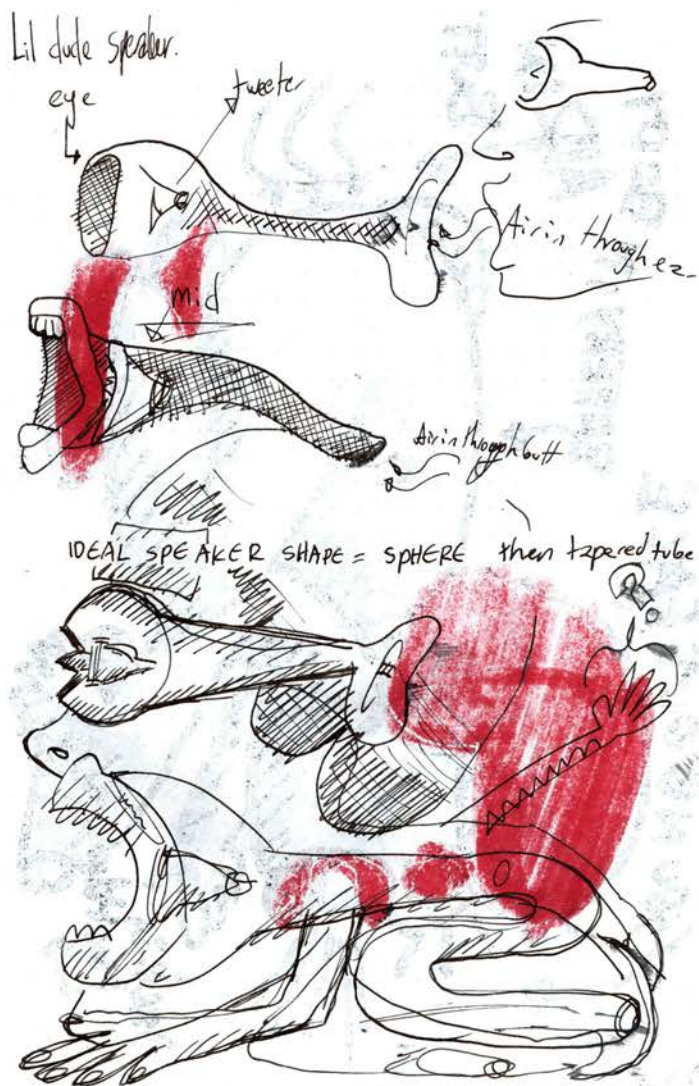


BELLOWING BOY



Designed and finished
by Ali Einoder - 2020



THE IDEA

Most of the objects we use in every day life are underwhelming. Simple shapes, solid colours, forgettable.

The items we have in our houses could look like a multitude of different things like lions or elephants or trees or planets or spaceships or anything really but most people instead settle for things that at a minimum look 95% like every other object of that function. Why is that?

Is it that we are afraid of the stimulation? Or maybe we are just afraid of taking the chance to see the world as a child would, with the surreal "what-if" leaps of imagination and the embracement of pareidolia.

Maybe it is too drastic a leap to change our environment into one in the vein of The Flintstones or Beauty and the Beast but it should be possible to sneak a little bit of that enjoyment into our lives.

A simple start is an object that is brought in and out of our attention by nature of its function. An item that is easy to anthropomorphize and one that can indulge this experiment without negating its usefulness.

This object is the speaker. An object that has become an almost necessity in modern life and has replaced a function that was once an intrinsic human service.

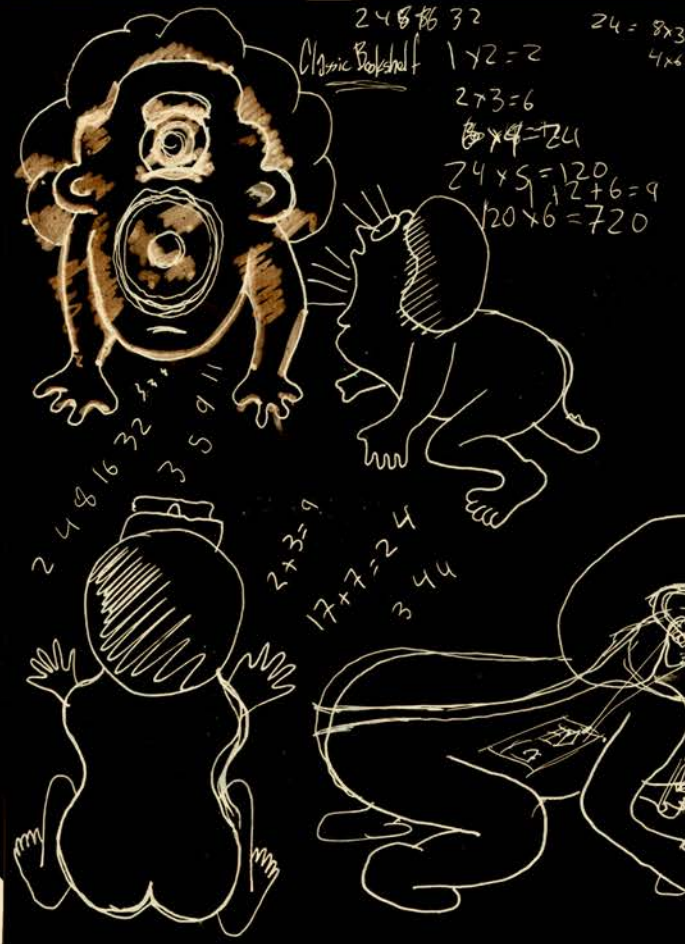
Why not return to listening to stories and songs and lectures from the mouth of an abstract humanoid.

Why not stare into his perfectly circular mouth and exotropic eyes?

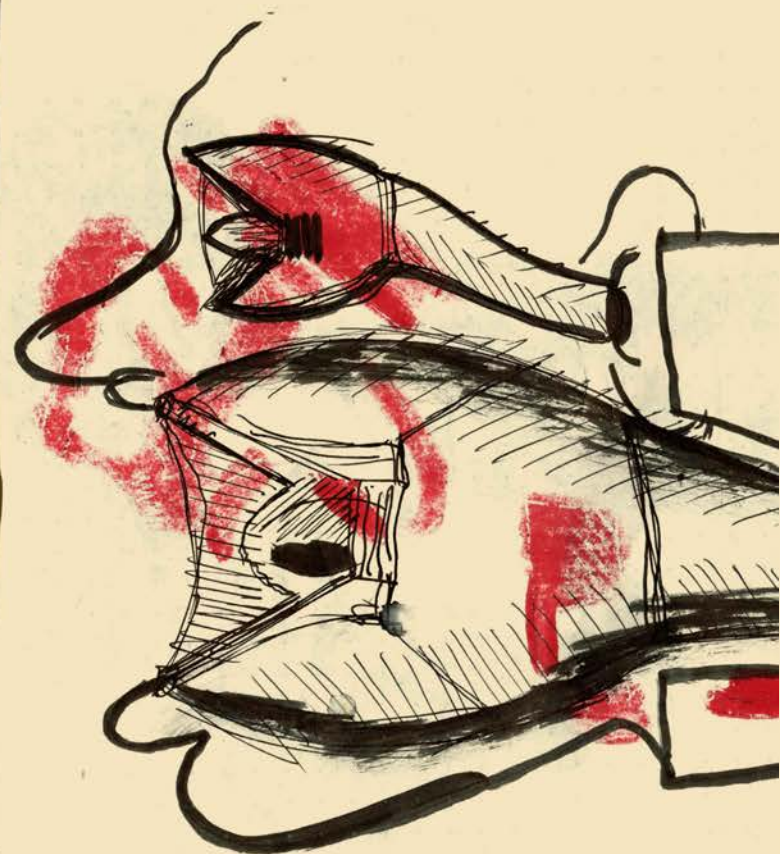
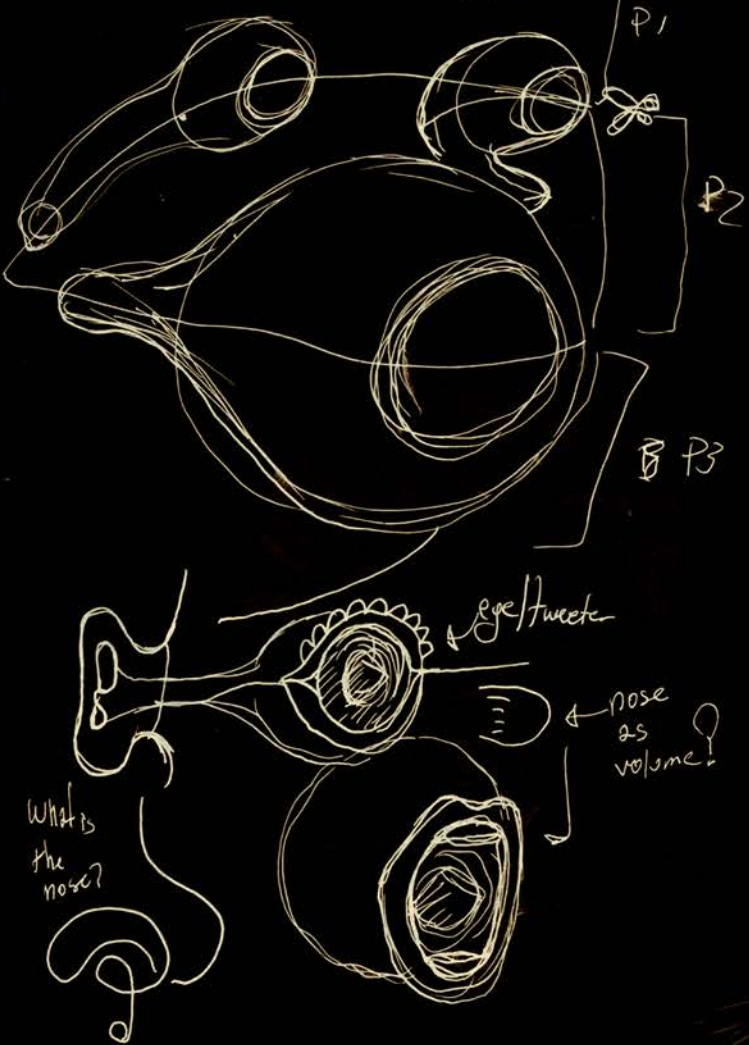
And while we are getting funky, we might as well Jackson Pollock him in paint, or graffiti him in Keith Haring doodles. This is an object that isn't planning to blend in like your nervous friend but is instead demanding attention.

You want every guest to be fixated, shocked, surprised and utterly exhilarated at the beautiful absurdity of this seemingly grotesque yet tantalizing figure. This boy crouching like the Sphinx, propelling whatever track or podcast you have been throttling lately right into the faces of its audience.

This will be the Yelling Boy, the boy who dares to change the world, bringing back blatant aestheticism and reference for the mere sake of it. A functional art-piece that can't help but stimulate you.



Will need to be in at least 3 pieces



THE DESIGN

To create a functional and heavily anthropomorphic speaker, several conditions need to be met for it to be considered a success. Firstly, it must resemble a humanoid creature, secondly it needs to have uncompromised audio quality and projection, thirdly it needs to be effortlessly interacted with.

An additional condition needs to be met considering I - the creator, need to make a prototype, This condition is that it is plausible to make using resources and finances I am easily able to access. These resources include a 3D printing facility, approximately 200 dollars, some paint I found in a parking lot, an audio driver able to be bought online and shipped within a week and the circuitry of a Google Home Mini I dissected out of curiosity.

The basics of the design were almost intuitive, instantaneously coming to me as I decided on the theme. The creature speaker was to bellow the sound out of its mouth and it was to either stand on its feet or crouch.

To be affordable and easy to print, the design needed to fit within an 8L cube, with no dimension greater than 20cm. It also needs to have the smallest weight possible without compromising any of the previously stated requirements.

To achieve this, the creature was to be mostly hollow, with walls of approximately 3mm thick. The more spherical main body seemed particularly useful as it was the most efficient use of space. This shape also allows for one other function that - especially for prototyping and development - is needed: The ability to open the device easily and access the inside. By having a large central body, a subsection of it can be removed to show all extra components inside such as the circuitry and the audio driver.

The last factor in the initial design was the size of the audio driver and circuitry that will be housed within. A driver that I have been needing to test out was the Dayton Audio ps95 which is a 3-1/2" mid range driver, perfect for this project. The circuitry was, as stated previously, that of a Google home Mini.



MODEL VERSION 1

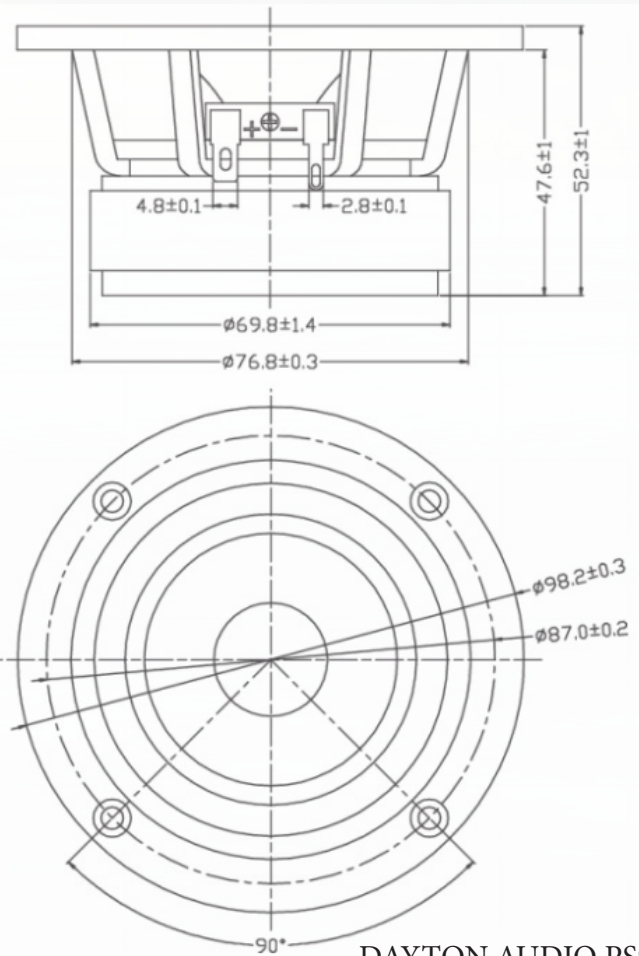
ACTUALLY MAKING IT / JAMMING

The first step in the 3D design was to map out the mounting of the audio driver (see image right), as it is an independent variable that the other structures must be designed around. The second step was to create a spherical chamber around the driver to aid acoustics.

The first version of the speaker (see images right) was created by adding the aesthetic features onto the outside of the central sphere. Muscular legs were added to the back and abstract skinny legs were attached to the front. Wings were added to the back both to visually balance the weight and also to make this creature resemble some sort of abstract cherub. The features were created cartoonishly and playfully, with the inclusion of solid pectoral muscles and 6-pack as well as small genitals reminiscent of Romantic and Classical sculpture.

As to allow airflow for the audio driver, an outlet was created in the rectum of the creature which also aids in the immature humour of its form.

This design, though meeting many



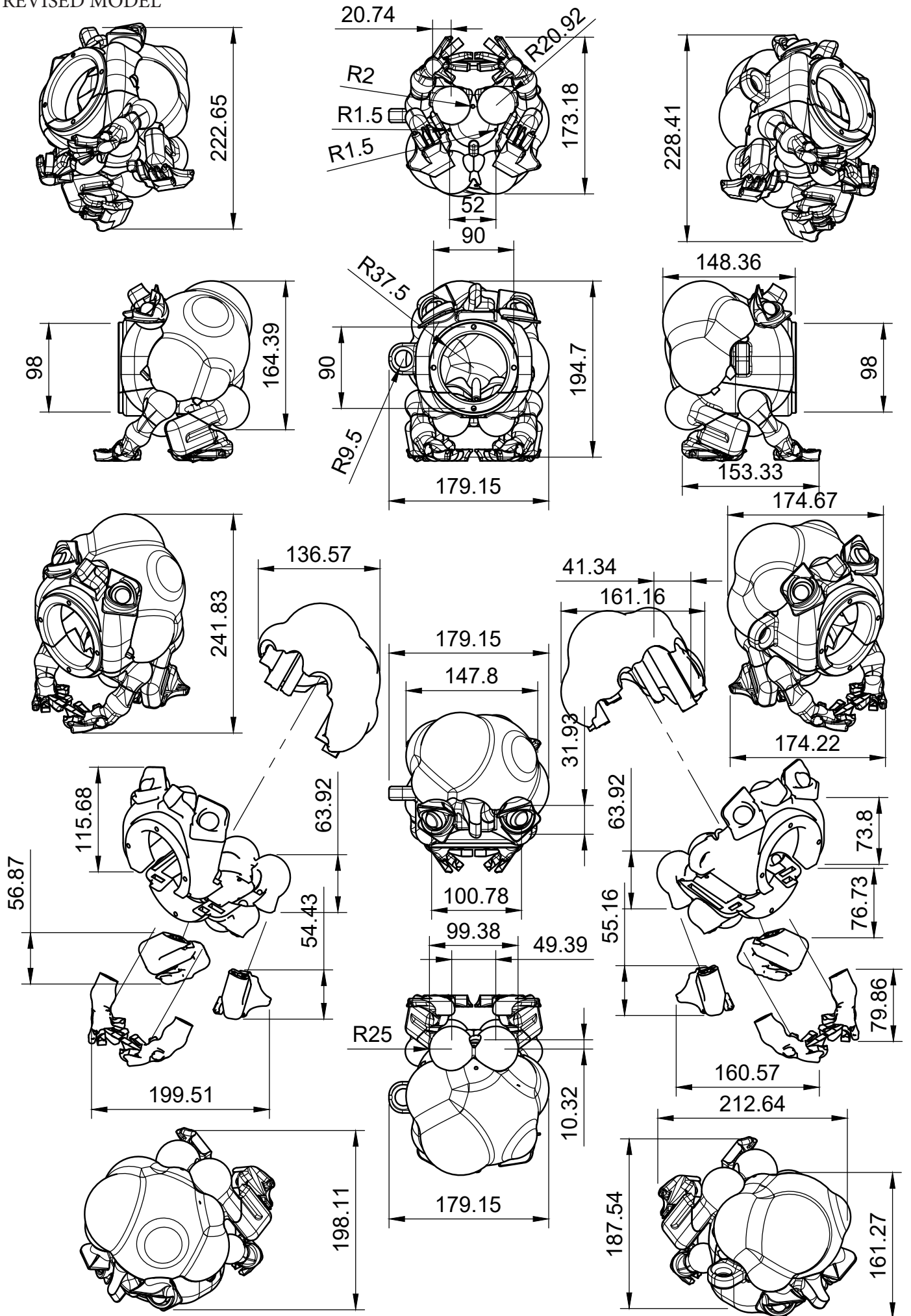
DAYTON AUDIO PS95

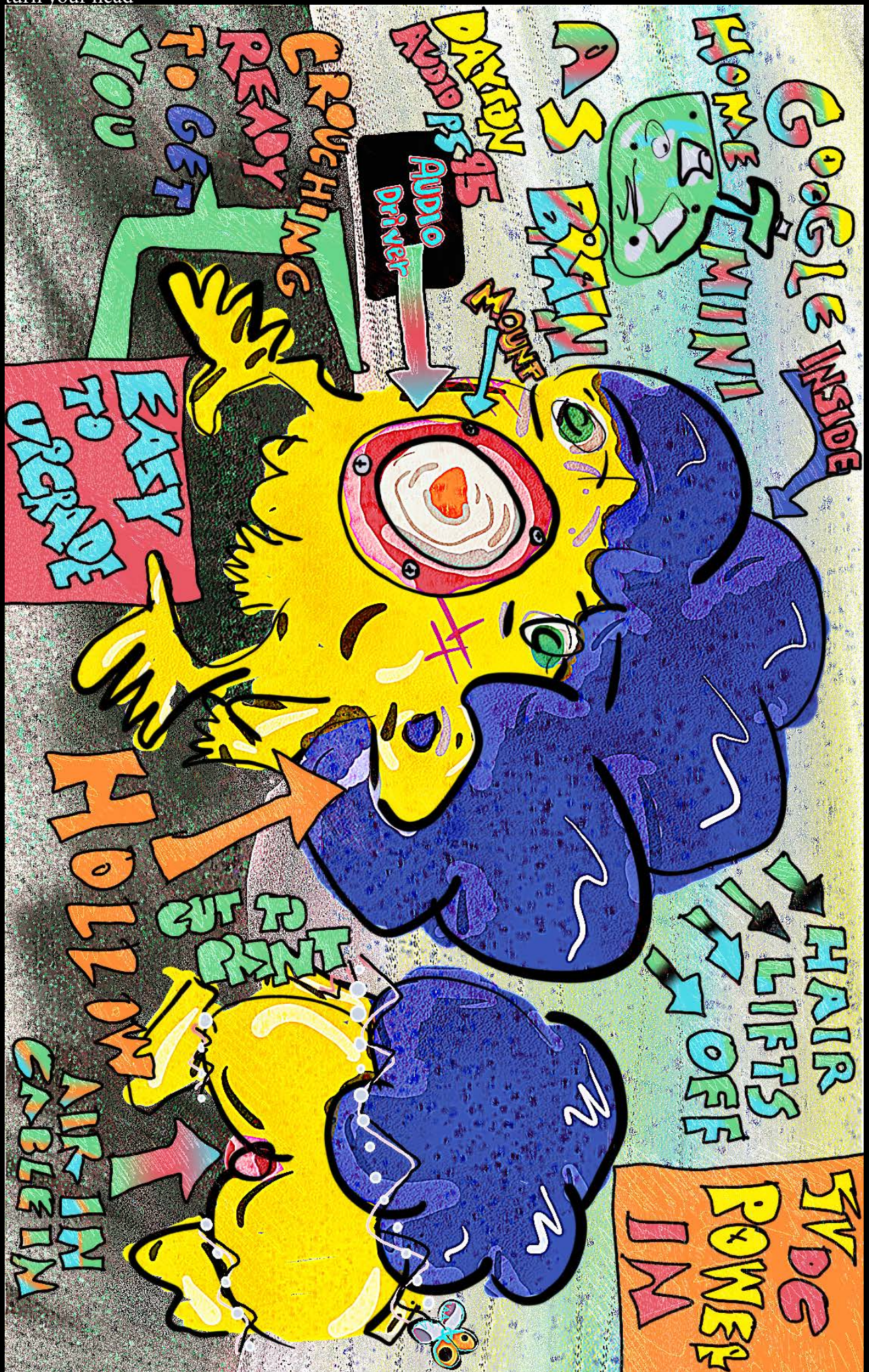
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of the requirements, failed one quite vital necessity; the size was far too large. The inefficient adding of solid bodies to the outside of the chamber created a large addition in weight which in turn lead to the cost of printing becoming far too high. Also, the size of this device was nearing 30cm in each dimension which is way too big.

Thus a new design was needed. This new design was to be more compact, with the aesthetic features being more integrated.

REVISED MODEL







This new design maintained the inner chamber being mostly spherical, however the top half was now a bubbly afro instead of solid spiky hair and wings. Whilst still being aesthetically grounded it is able to double as a removable lid. The legs were made smaller than that of the previous design but also staying relatively in proportion to the new form. The stance is that of a frog, with the mouth facing the flies. The eyes staying exotropic and upwards facing to accentuate the playful absurdity





of the design. To minimise cost, the central body has thin walls and detaches from the limbs and hair and is able to be split in two. This splitting of components greatly reduces the support structures needed during the printing process and thus reduces the cost dramatically.

The overall size of this design was reduced to a fraction of the previous with greater efficiency of the structures and a more unified form.





Removing the hair exposes the mounting area for the Google Home Mini circuitry (see above) and the Dayton Audio ps95 (see below). The hair itself contains finger like integrated dowls that insert themselves into the main body. This holds itself in place at an angle of forty five

degrees to ensure stability, reliability and the extra ability to hold the rest of the construction in place. This function is a key example of the multifaceted functioning of the components. The hair in total, works as a buttress for the face, helping to support the weight of the audio driver whilst in addition aiding





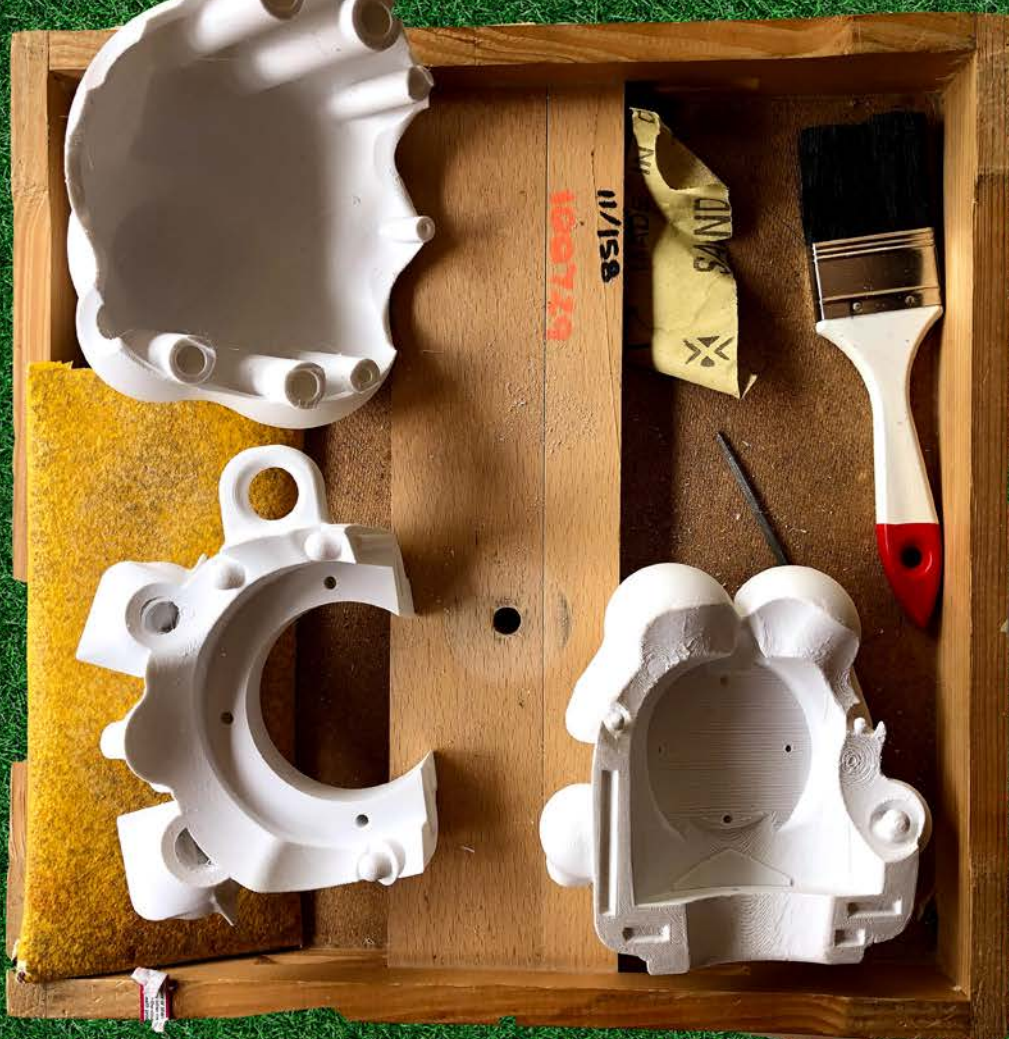
the acoustic qualities of the speaker and providing access to the internals.

The face of the Bellowing Boy also removes (see bellow). This is a one time function merely implemented to aid the printing process and greatly reducing cost by decreasing the total amount of

supportive filament needed. This component, as well as the limbs, are attached with some super heavy duty smelly quick drying super glue. To aid the attachment, the joints were given teeth and slots, increasing the total surface area of the connection and providing protection from greater angular stress.



COMPONENTS POST-PRINT



POST-PRINT PROCESSING

With the assembly printed and delivered, the most tedious process begins. Each part needs to be sanded, the wrinkles and contours of the prints must be smoothed into a slippery surface. Each of the joining components need special care to connect to their full extent. The components need to be tested and adjusted, in this case using a file, brute force and fire. Heating the plastic allows for speedy adjustment of components that otherwise would be impossible to reform.

The parts sans the hair were glued together using super glue and high heat while being pressed together with force.

After joining, the components were let to rest for several hours so I could lie down and watch some Netflix.



THE PAINTING THE PAINTING THE PAINTING

The painting was a key component to the overall design as I find an incessant need to cover my creations in paint. This, paired with a liking of bold primaries and rough abstract composition leads to a personalisation and giving of character to the product.

The reclaimed, post-trash paints demanded to be poured and dripped with a Pollock spirit, covering the Bellowing Boy like multicoloured pigeon poop in a surreal Italian Piazza. This aggressive painting serves as a practical first composition as it allows for further mutations through graffiti and wear and tear.

An additional, hidden benefit of the heavy paint is that it further covers the lumpy, bumpy washing-board curves left by the 3D printer as well, as the burnt and glued joints.

The final step of the painting required burning the paint, forcing it to bubble and swell. This causes the different colours to scatter throughout each other without mixing, it also creates a weathering, stripping the product of its sterile newness and adding depth to the colours.



THE RESULT

Due to the precision of measurement in the 3D modeling process as well as the heavy effort in post-processing and adjustment, the parts all fit together effortlessly. The circuit board can attach to the mounting holes of the base of the inside. The hair is easily removed and connected whilst still holding tightly. The Audio Driver mounts perfectly and tightly, not moving a micrometer even when thumping hard bass.

The weight is high enough to suggest quality but it is light enough to easily move around. The Google Home Mini brain is able to be communicated with, even with its encapsulation and its close proximity to the speaker. Its cord effortlessly passes through the rectum and can be attached and reattached if the device is opened.

In time when the paint dries completely, additional detailing and graffiti will be applied. The circuitry may be replaced with a more conventional Bluetooth amplifier. But for the time being the Bellowing Boy crouches proud. Gleefully scaring unsuspecting interlopers and small dogs that come sniffing.



